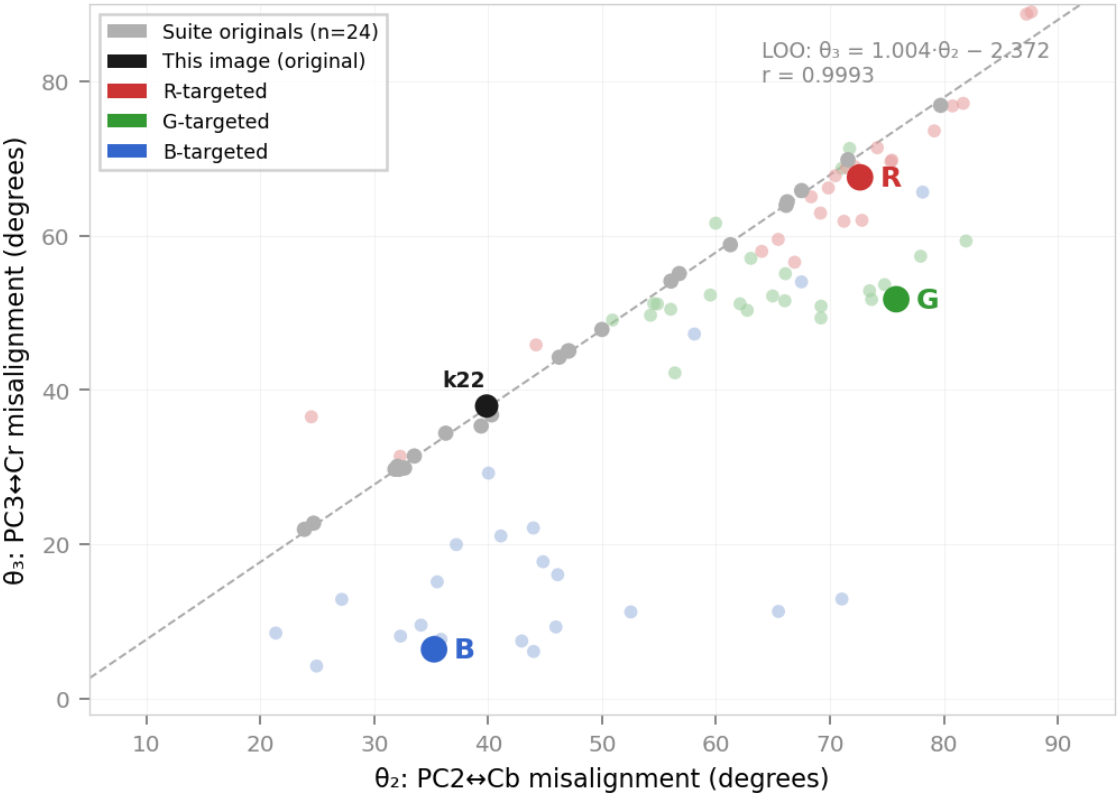


Compression Profile: KODIM22 768x512 | 24-bit RGB | PNG (lossless)

barn and pond (KINSA Photo Contest) | Weakly One-Dimensional (PC1 = 89.76%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8714	0.8001	0.8752	0.849	39.86°	37.93°	+0.28°
R-targeted	0.1021	0.0943	0.9918	0.396	72.64°	67.59°	-2.97°
G-targeted	-0.0724	0.9531	-0.0909	0.372	75.82°	51.78°	-21.97°
B-targeted	0.9399	0.0545	0.0465	0.347	35.24°	6.38°	-26.63°

R-targeted isolates red (R-G, R-B < 0.10) while preserving G-B coupling at 0.992.
G-targeted isolates green (R-G, G-B < 0.09) while preserving R-B at 0.953.
B-targeted preserves R-G at 0.940 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM22 768x512 | 24-bit RGB | PNG (lossless)

barn and pond (KINSA Photo Contest) | Weakly One-Dimensional (PC1 = 89.76%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

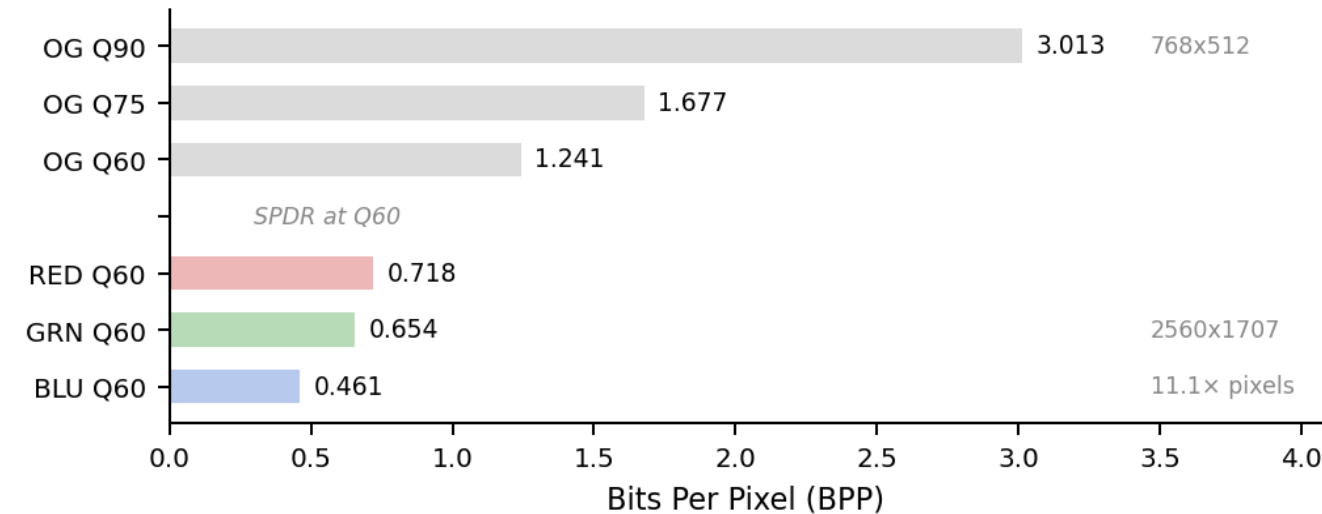
ORIGINAL (kodim22)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	685.5 KB	14.282	393,216
JPEG Q90	JPEG	768x512	144.6 KB	3.013	393,216
JPEG Q75	JPEG	768x512	80.5 KB	1.677	393,216
JPEG Q60	JPEG	768x512	59.6 KB	1.241	393,216

REDISTRIBUTED (kodim22 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	383.0 KB	0.718	4,369,920
GRN Q60	JPEG	2560x1707	348.7 KB	0.654	4,369,920
BLU Q60	JPEG	2560x1707	245.9 KB	0.461	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM22 768x512 | 24-bit RGB | PNG (lossless)

barn and pond (KINSA Photo Contest) | Weakly One-Dimensional (PC1 = 89.76%)

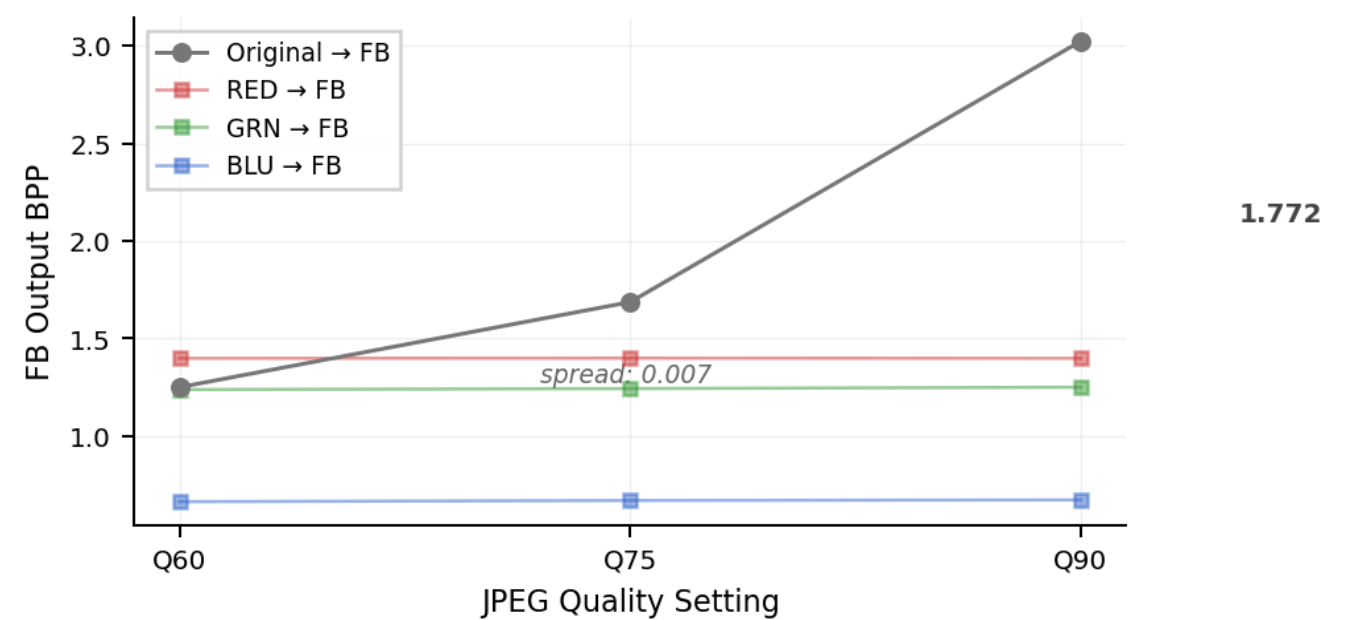
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	2.697	—
R-targeted → FB	2560x1707	2048x1366	1.399	48.1%
G-targeted → FB	2560x1707	2048x1366	1.251	53.6%
B-targeted → FB	2560x1707	2048x1366	0.674	75.0%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.443	46.5%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8491 vs 0.849 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.772 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.007 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim22

Best BPP (Q60, direct)	0.461	2560x1707, 4,369,920 px
Best BPP through FB	0.674	2048x1366 output
vs OG Q90 → FB	77.7%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216